

# SATHWIK H NAIK

| (240) 438-0411 | [sathwikhnaik@gmail.com](mailto:sathwikhnaik@gmail.com) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) |

## PROFESSIONAL SUMMARY

Data scientist with hands-on forecasting, causal inference, and experimental design experience at a Fortune 50 healthcare organization. Completed M.S. in Data Science at UMD in 2026, backed by a banking analytics capstone measuring a marketing campaign's causal impact.

## EDUCATION

University of Maryland, College Park

2024 – 2026

M.S. in Data Science

GPA 3.81/4.0

Coursework Statistical Inference, Deep Learning, Data Modeling, Data Visualization, Experimental Design, Advanced Time Series

Reva University, India

2019 – 2023

B.Tech, Computer Science and Engineering

GPA 3.55/4.0

## WORK EXPERIENCE

Elevance Health

Aug 2023 – Aug 2024

Software Engineer

- Designed reusable KPI structures, translating operational requirements into consistent, self-serve cross-team metrics
- Developed ETL workflows to source, clean, and validate healthcare data, improving quality for stakeholder reports
- Automated recurring data-preparation and reporting workflows, eliminating 20+ hours of manual weekly effort
- Built and maintained 10+ Power BI dashboards with DAX measures for multiple internal teams
- Documented report logic, metric definitions, and validation rules enabling governed self-service reporting

## PROJECTS

Banking Liquidity Forecast Lab ([GitHub](#))

Tech Stack: Python, LightGBM, statsmodels, FastAPI, Docker Compose

- Measured a marketing campaign's causal lift using interrupted time series and difference-in-differences
- Recovered a known 12% campaign lift within 0.4 percentage points using two causal estimators
- Achieved 9.48% WAPE forecasting daily cash outflow with Holt-Winters under six-fold walk-forward validation

RAG Evaluation Lab ("Retrieval Quality Lab") ([GitHub](#))

Tech Stack: Python, ChromaDB, Ollama, RAGAS, LangChain, sentence-transformers

- Evaluated generation quality using EM, F1, a custom grounding score, and RAGAS relevancy
- Diagnosed generation, not retrieval, as the system bottleneck despite 95% gold-doc accuracy
- Achieved 85.3% Recall@1 with hybrid retrieval plus cross-encoder reranking, up from 69.6% BM25

SaaS Subscription Churn & Revenue Analytics Platform ([GitHub](#))

Tech Stack: DuckDB, pandas, PostgreSQL, scikit-learn, SHAP

- Trained a Gradient Boosting churn model and explained predictions per-customer using SHAP
- Reconstructed point-in-time customer state using pandas merge\_asof across 8,000 synthetic customer accounts
- Derived 15+ metrics tables including NRR, GRR, and ARPA from raw subscription events

## TECHNICAL SKILLS

- **Statistics & Methods** Statistical Inference, Experimental Design, A/B Testing, Causal Inference (ITS, DiD), Time-Series Forecasting
- **ML & AI** PyTorch, scikit-learn, LightGBM, Gradient Boosting, SHAP, GANs, Deep Learning
- **LLM & RAG Engineering** RAG Evaluation, Hybrid Search, ChromaDB, RAGAS, Recall@k/MRR
- **Data & SQL** SQL, DuckDB, PostgreSQL, ETL, Data Pipelines, Snowflake
- **Programming & Tools** Python, pandas, R, Git, Docker, FastAPI, CI/CD